Example 8.5

Let $f, g : (a,b) \to \mathbb{R}$ be differentiable functions. Show that $d(fg) = fdg + gdf$.

Solution

Let $f, g : (a,b) \to \mathbb{R}$ be differentiable functions and $h(x) = f(x)g(x)$. Then h, being a product of differentiable functions, is differentiable on (a,b). So by definition $dh = h'(x)dx$. Now by using product rule we have $h'(x) = f(x)g'(x) + f'(x)g(x)$.

Thus $$dh = h'(x)dx = (f(x)g'(x) + f'(x)g(x))dx = f(x)g'(x)dx + f'(x)g(x)dx$$
$$= f(x)dg + g(x)df = fdg + gdf$$

Example 8.6

Let $g(x) = x^2 + \sin x$. Calculate the differential dg.

Solution

Note that g is differentiable and $g'(x) = 2x + \cos x$.

Thus $dg = (2x + \cos x)dx$.

Example 8.7

If the radius of a sphere, with radius 10 cm, has to decrease by 0.1 cm, approximately how much will its volume decrease?

Solution

We know that volume of a sphere is given by $V = \frac{4}{3}\pi r^3$, where $r > 0$ is the radius. So the differential $dV = 4\pi r^2 dr$ and hence

$$\Delta V \approx dV = 4\pi(10)^2(9.9\text{-}10)\,\text{cm}^3$$
$$= 4\pi 10^2(-0.1)\,\text{cm}^3$$
$$= -40\pi\,\text{cm}^3.$$

Note that we have used $dr = (9.9 - 10)$ cm, because radius decreases from 10 to 9.9. Again the negative sign in the answer indicates that the volume of the sphere decreases about 40π cm^3.

EXERCISE 8.2

1. Find differential dy for each of the following functions :

   (i) $y = \dfrac{(1-2x)^3}{3-4x}$ (ii) $y = (3 + \sin(2x))^{2/3}$ (iii) $y = e^{x^2-5x+7}\cos(x^2-1)$

2. Find df for $f(x) = x^2 + 3x$ and evaluate it for

   (i) $x = 2$ and $dx = 0.1$ (ii) $x = 3$ and $dx = 0.02$

3. Find Δf and df for the function f for the indicated values of $x, \Delta x$ and compare

   (i) $f(x) = x^3 - 2x^2; x = 2,\ \Delta x = dx = 0.5$

   (ii) $f(x) = x^2 + 2x + 3;\ x = -0.5,\ \Delta x = dx = 0.1$

4. Assuming $\log_{10} e = 0.4343$, find an approximate value of $\log_{10} 1003$.

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